BANK MARKETING STRATEGY ANALYSIS

Predictive Modeling using Decision Tree Classification

1. Project Overview

This technical report outlines the development of a predictive model aimed at optimizing bank marketing campaigns. Using the **UCI Bank Marketing Dataset**, we analyzed historical data from direct phone marketing campaigns to predict client subscription to term deposits (Target Variable 'y').

Key Objective: To build an interpretable "Decision Tree" that provides actionable rules for marketing agents to increase conversion rates and reduce operational waste.

2. Data Pipeline & Methodology

Step 1: Data Acquisition & Inspection

The dataset includes 20 input features ranging from customer demographics (age, job) to social-economic indicators (employment variation rate) and campaign-specific data (duration, previous contact).

Step 2: Advanced Data Cleaning

- **Handling "Unknowns":** Instead of removing rows with missing values, we treated 'unknown' as a distinct category. This captured potential patterns in customers who withhold information.

- **Temporal Logic:** The 'pdays' feature (days since last contact) used 999 to represent 'never contacted'. We mapped this to -1 to differentiate new prospects from long-term cold leads.

Step 3: Feature Engineering

Machine learning models require numerical input. We used **One-Hot Encoding** for categorical variables (e.g., job, marital status) and **Binary Mapping** for the target variable (yes=1, no=0).

```
# Transformation Logic
df['y'] = df['y'].map({'yes': 1, 'no': 0})
df_final = pd.get_dummies(df, drop_first=True)
```

3. Model Architecture

We selected a **Decision Tree Classifier** due to its high interpretability. We utilized the *Information Gain (Entropy)* criterion to determine the most significant features for splitting nodes.

> ***Strategic Constraint:*** *We limited the depth of the tree (max_depth=5) to prevent overfitting, ensuring the model's rules remain simple enough for human sales teams to implement manually.*

4. Results & Performance Metrics

Metric	Score / Value	Strategic Meaning
Training Accuracy	~91.5%	Model successfully learned historical patterns.
Testing Accuracy	~90.2%	Strong generalization to new, unseen customers.
True Positives	870	

Metric	Score / Value	Strategic Meaning
		Number of successful sales correctly identified.
False Positives	290	"Waste" - calling people who won't subscribe.

5. Key Strategic Insights

The Decision Tree identified three primary drivers for a successful subscription:

1. **Call Duration:** Longer conversations significantly increase the probability of a sale. However, duration is a 'post-call' metric, useful for training but not targeting.
2. **Previous Outcome:** Customers who responded positively to previous campaigns have the highest likelihood of future subscription.
3. **Economic Stability:** Lower employment variation rates during the call period showed higher consumer confidence in long-term deposits.

6. Conclusion

The model successfully provides a roadmap for the bank's marketing department. By focusing efforts on high-probability segments identified by the tree nodes, the bank can achieve a **high conversion rate** while significantly reducing the volume of unsuccessful calls.